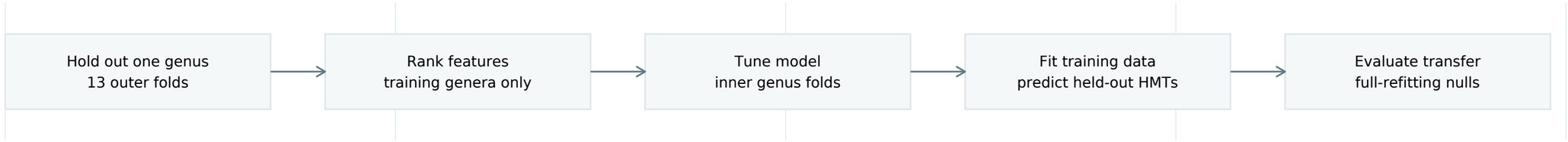
A

Genus-held-out prediction design

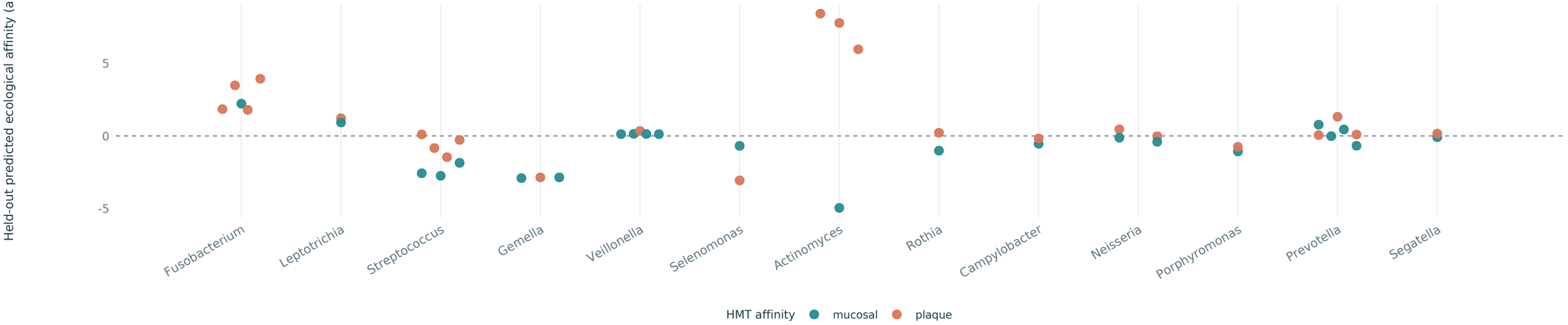
Held-out genera are excluded from feature ranking, tuning, scaling and fitting



B

Predictions for all 47 held-out HMTs

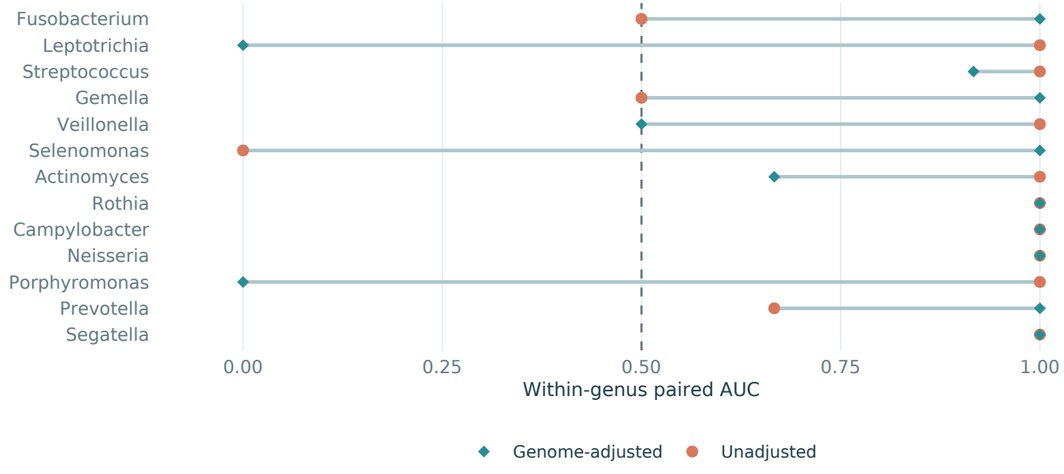
Source-family-balanced outcome; each point is one taxon



C

Transfer varies across genera

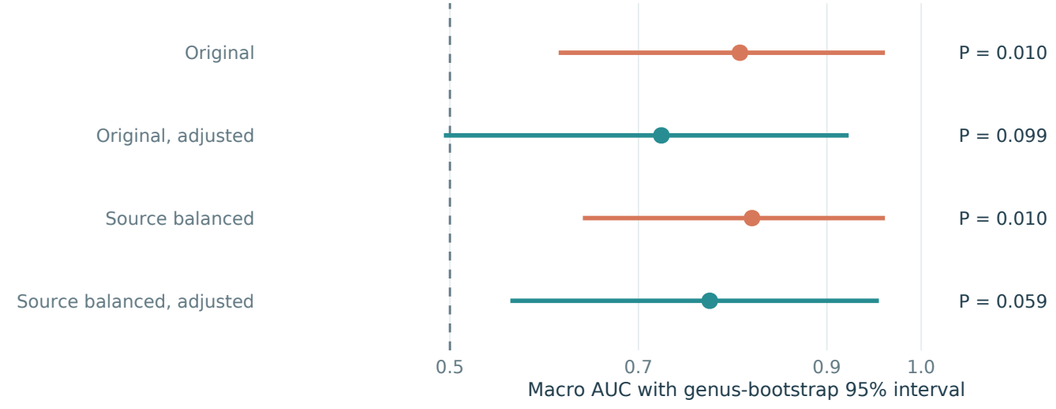
Source-balanced models; dashed line: chance



D

Prediction and covariate sensitivity

Full-refitting permutation P values



Unadjusted nulls: 200 complete refits per outcome. Genome-adjusted nulls: 100 refits. HMTs, rather than individual genomes, are the prediction units.